involves patterns and predictions.

Accuracy is of paramount importance: if you thought a filter with 99 per cent accuracy is good enough, think again. You'd rather wade through a hundred spam e-mails a day than lose that important e-mail from your boss, or worse still have your boss classified as a spammer and blocked without your knowledge. Spam classifiers can botch things in two ways: classifying non-spam as spam, called false positives, and spam as non-spam, called false negatives. The latter is certainly preferable to the former; but we wouldn't want too many of either. Another problem is that people tend to rely on and trust their spam blocker too much, and to not double-check those marked as spam, for false positives. Accuracies, therefore, need to reach something like 99.99 per cent. Bayesian spam blockers learn from what you tell them, improving their accuracy every day.

Consider some of the things that make Bayesian filters more effective than traditional techniques:

These filters see causal connections. So, if the word 'debt' is evidence of spam, but the phrase 'get out of' is not, Bayesian filters would notice, all by themselves, that e-mails with both these in them—as in 'get out of debt'—are spam.

They evolve with spam. As spammers' techniques change the filter adapts. Even if the spammer starts saying "Get 0ut of debt"—with a zero instead of an 'o'—the filter would notice, because of the pattern in the other letters.

Since they are aware of causal connections, both 'good' words as well as 'bad' words contribute to the decision.

Eric Horvitz
Senior Researcher,
Microsoft Research

> "I believe that probability is at the **foundation** of any **intelligence** in an uncertain world where you can't **know** everything"

'Gleeful' or 'Bayesian' hardly occur in spam, and this makes as much of a difference to the learning process as words such as 'debt' or 'xxx.'

Two open-source spam killers are SpamBayes and PopFile. SpamBayes is different; in addition to 'spam' and 'non-spam,' it can assign an 'unsure' classification too—so there's virtually no risk of losing a good e-mail. PopFile uses a Naïve Bayes classifier (NBC). An NBC operates taking only two factors into account: the number of spams you receive, and the ratio of the probability of any word in the dictionary occurring in spam versus that of it occurring in non-spam. If 'debt' occurs with a 40 per cent chance in spam, and with a 2 per cent chance in non-spam, this ratio is 20. Surprisingly, even using just these two factors, NBCs are very effective.

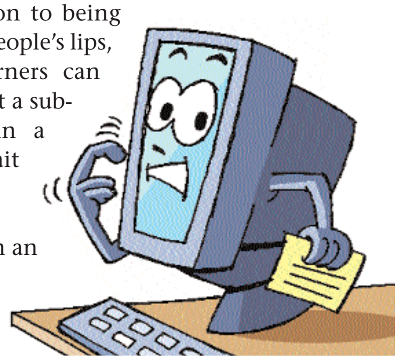
As an example of how effective Bayesian searching and classification is, consider these: Microsoft plans to incorporate Bayesian in Longhorn for search and other functions. Netscape 7.1 includes Bayesian anti-spam. AOL is now using Bayesian to kill spam. Jim Nail, a Forrester Research analyst, believes this is a more sophisticated approach than others: "The tools in the 9.0 release could be a major improvement… I don't think spam has become a game-loser for AOL just yet." And Paul Graham, designer of the Arc language and a major crusader in the spam battle, says: "Of all the approaches to fighting spam, from software to laws, I believe Bayesian filtering will be the single most effective."

Bayesian everywhere

In biology, researchers are using Bayesian to determine previously unsuspected correlations between symptoms and diseases. Also, searching through gene data, correlations between genes and cell behaviour have been found. Such correlations are useful in cancer research, for example.

Bayesian prediction naturally finds wide application in the fields of finance and commerce. Stocks and shares are unpredictable, and it's intuitive that there are hidden patterns at work, which makes the stock market an ideal candidate for Bayesian intervention. A real-world example of a BBN at work in finance is Bayes-Credit, which is used for predicting the risk that a corporate will crash within a given timeframe.

In addition to being able to read people's lips, Bayesian learners can also find what a subtle change in a person's gait can mean. Certain changes mean an impending stroke,



He's Probably In The Shower

Microsoft's Notification Platform (NP), part of .NET, employs Bayesian techniques. The NP will enable cell phones and computers to automatically filter messages, schedule meetings without human intervention, and find the best strategy to contact someone. An element of the NP called Coördinate can gather data from keyboards, personal calendars, sensor cameras, etc, to create an entire 'person profile.' In this profile, one can expect to see the person's schedules, the time and duration of work hours and lunch hours, what kind of messages he sends and receives, and so on. The NP then uses this profile to arrange for the best possible time and place to deliver the information, or notification.

Say you want to contact someone: you get onto the NP, and declare that you want to contact Mr X. The NP will first identify you, then Mr X, decide whether he's busy or not, what kind of communication he'll prefer at this time of the day, etc. It'll then advise you of the best mode and time to contact Mr X.

If the NP turns out to be a success, 'context servers' could be next: electronic butlers that'll help people organise their information and their schedules.